L-3

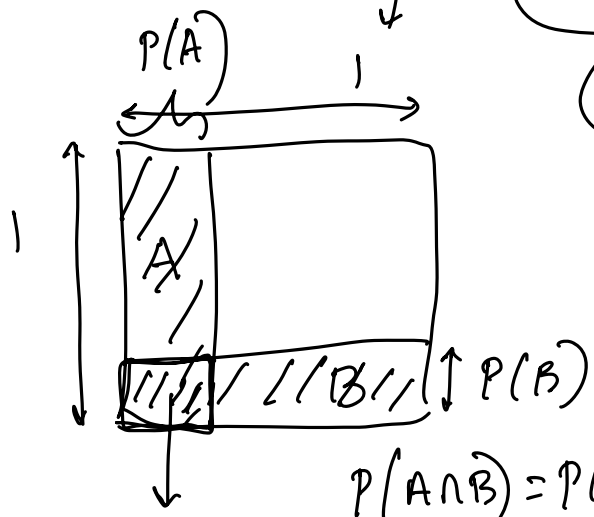
A, B are independent if

$$P(A \cap B) = P(A)P(B)$$

2^n

$$\sum_{r=2}^{n} {}^nC_r \rightarrow$$

$$2^n - n - 1 \rightarrow$$



$$P(A \cap B) = P(A)P(B)$$

Remark: Independence $\Leftrightarrow$ A & B are orthogonal to each other

Newton-Pepy's Problem (1693) Roll a fair dice. Which of these is most likely?

Truth

(A) at least one 6 with rolling 6 dice.

(B) at least 2 6's " " 12 dice

(C) " " 3 6's " " 18 dice

Ans. (A)

$$P(A) = 1 - \underbrace{\left(\frac{5}{6}\right)\left(\frac{5}{6}\right)\cdots\left(\frac{5}{6}\right)}_{6 \text{ times}}$$

$$1 - \left(\frac{5}{6}\right)^6 \approx 0.665$$

$$P(B) = 1 - \underbrace{\left(\frac{5}{6}\right)^{12}}_{\downarrow} - \underbrace{12\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^{11}}$$

$$\approx 0.619$$

$$P(C) = 1 - \underbrace{\sum_{k=0}^{2} {}^{18}C_k \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{18-k}}_{\downarrow}$$

$$\approx 0.597$$

Conditional Probability

Def^n: The conditional probability of A given B is

$$\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}}$$

assuming $P(B) > 0$

Ex. Interpret using Venn diagrams

Remark: If A & B are independent, then

$P(AB) \longleftarrow P(A \cap B) = P(A)P(B)$

& $\Rightarrow P(A|B) = P(A)$

Ex. $P(\text{both aces} \mid 1 \text{ of them is an ace})$

$P(\text{both aces} \mid 1 \text{ of them is an ace of spades})$

$$= \frac{P(\text{both aces } \& \ 1 \text{ of them is an ace})}{P(1 \text{ being an ace})}$$

also

$$= \frac{{}^4C_2 / {}^{52}C_2}{1 - {}^{48}C_2 / {}^{52}C_2} = 1/33$$

$P(\text{both aces} \mid 1 \text{ of them is an ace of spades})$

$$= \frac{3}{51} = \frac{1}{17}$$

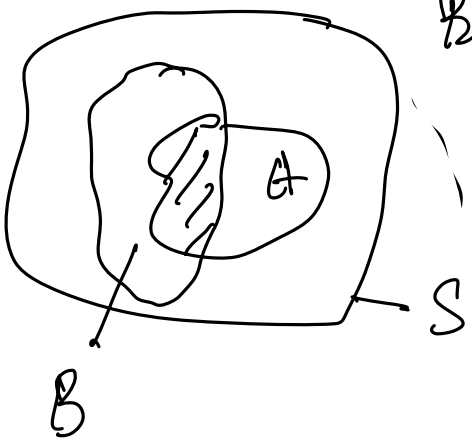
AS

1. Try simple & extreme cases
2. Break the problem into simpler problems

A & B event in S

Find $P(A)$

Break it into pieces using B



$$A = \underbrace{(A \cap B)} \cup \underbrace{(A \cap B^c)}$$

mutually exclusive

$$(A \cap B) \cap (A \cap B^c) = \emptyset$$

$$P(A) = P(A \cap B) + P(A \cap B^c) \longrightarrow (*)$$

By conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$\boxed{P(A) = P(A|B)\,P(B) + P(A|B^c)\,P(B^c)}$$

Total Probability theorem

$$P(A|B)\,P(B) = P(B|A)\,P(A)$$

$$\boxed{P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}}$$

Bayes rule

Def^n: The events $C_1, C_2, \ldots, C_n$ are called a 'Partition of A' if

(i) If $i \neq j$, $C_i \cap C_j = \emptyset$

(ii) $C_1 \cup C_2 \ldots \cup C_n = A$

Prop^n: If $C_1, C_2 \ldots, C_n$ is a partition of A.

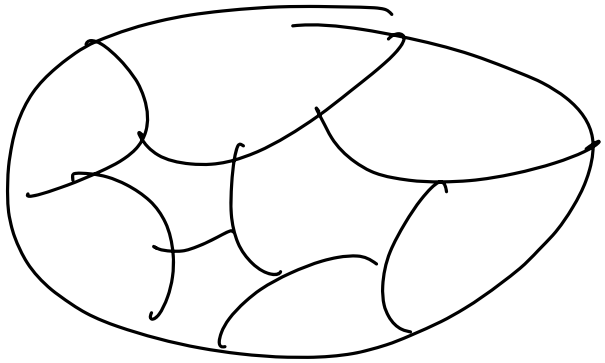
Then

$$P(A) = P(C_1) + P(C_2) + \cdots + P(C_n)$$

If $B_1, B_2 \ldots B_n$ is a partition of S

let A be a generic event in S.

then $A \cap B_1, A \cap B_2 \ldots, A \cap B_n$ is a partition of A



$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_n)$$

$$\boxed{P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)}$$

Generalized TPT

$$\left.\begin{aligned} P(A \cap B) &= P(A)P(B|A) \\ &= P(B)P(A|B) \end{aligned}\right\}$$

Thm

$$P(A_1 \cap A_2 \cap \dots \cap A_n)$$
$$= P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2)$$
$$P(A_4|A_1 \cap A_2 \cap A_3)$$
$$\dots P(A_n|A_1 \cap A_2 \cap \dots \cap A_{n-1})$$

Ex. Complete the pf.

$n!$ theorems

$$P(A_{10})\, P(A_1 \mid A_{10})\, P(A_2 \mid A_1 \cap A_{10}) \cdots\cdots$$

$$= P(A_3)\, P(A_1 \mid A_3) \cdots\cdots$$